

Phoenix-RTOS

Preparing the environment

Wiktor Daszczuk wiktor.daszczuk@pw.edu.pl

Waldemar Grabski waldemar.grabski@pw.edu.pl

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2 INSTALLING UBUNTU

2.1 INSTALLING AND CONFIGURING UBUNTU IN A VIRTUALBOX VIRTUAL MACHINE

We install Ubuntu in server version only with a text-based interface (no graphical interface) in a VirtualBox virtual machine.

2.1.1 Software version

VirtualBox version: 7.0.12 r159484

Operating system version: Ubuntu server 22.04.1 LTE <https://ubuntu.com/download/server>

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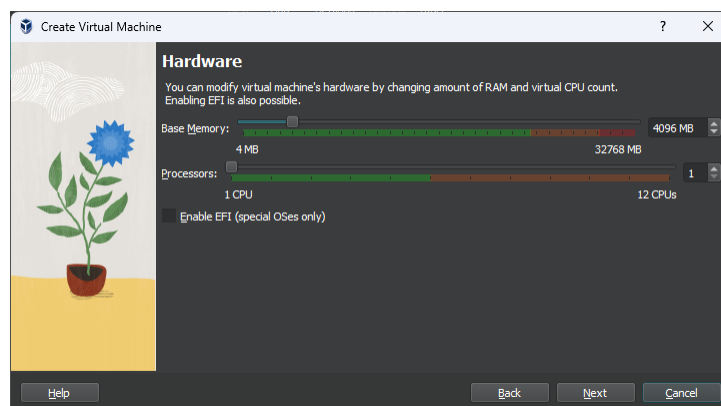
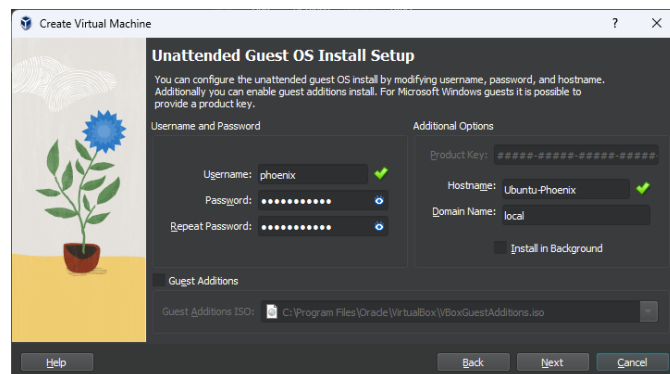
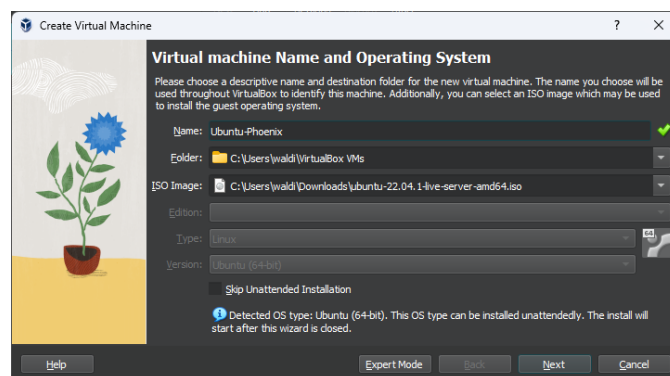
2.1.2 Virtual machine parameters

Virtual machine parameters:

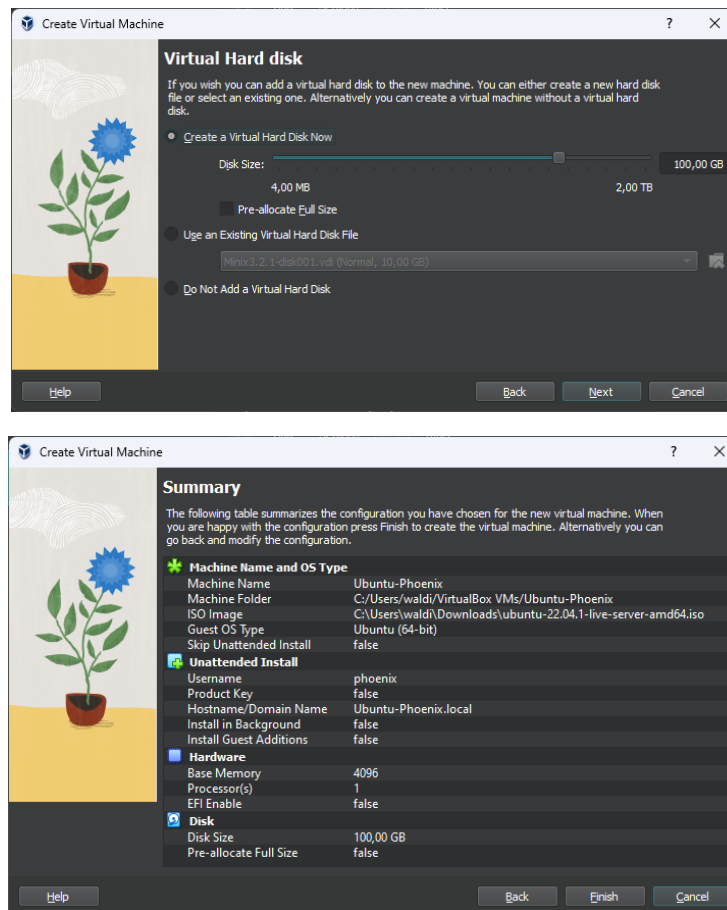
- Machine Name: Ubuntu-Phoenix,
- Domain: local,
- User: phoenix, password: phoenixrtos,
- Guest operating system: Linux, Ubuntu (64-bit),
- RAM: 4096MB (at 1 GB the tools do not build correctly - compiler errors),
- Disk: VDI, Dynamically allocated, 100GB,
- Connected ISO image: ubuntu-22.04.1-live-server-amd64.iso

Select the unattended installation.

When you select an iso image, the type (Linux) and Version (ubuntu (64bit)) are set.



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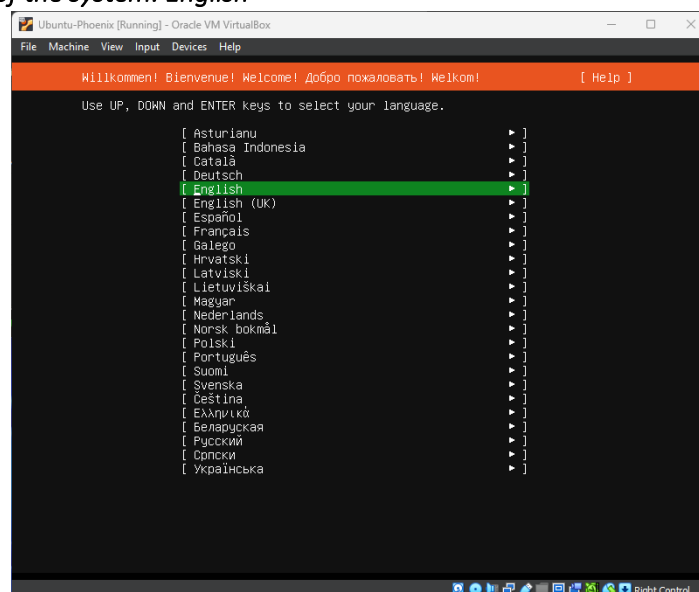


Once the settings are complete, an unattended installation of Ubuntu is launched.

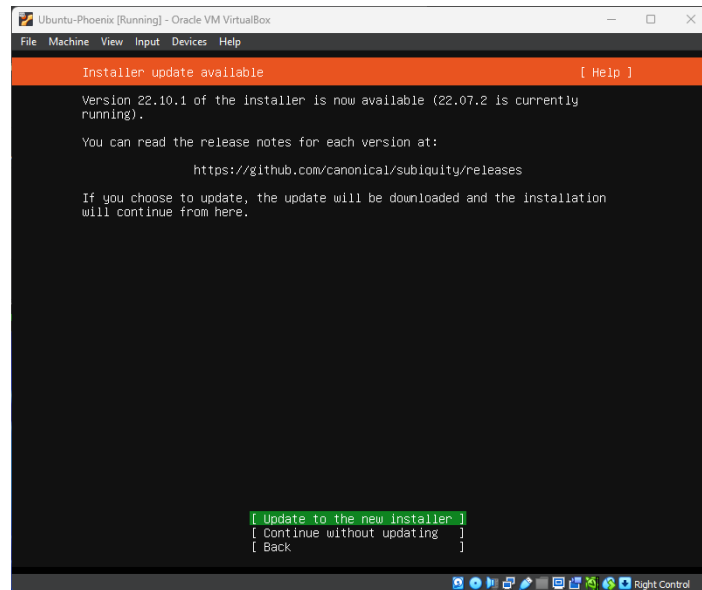
2.1.3 Ubuntu operating system Installation

We install the text version and *the OpenSSH* server, without any additional components.

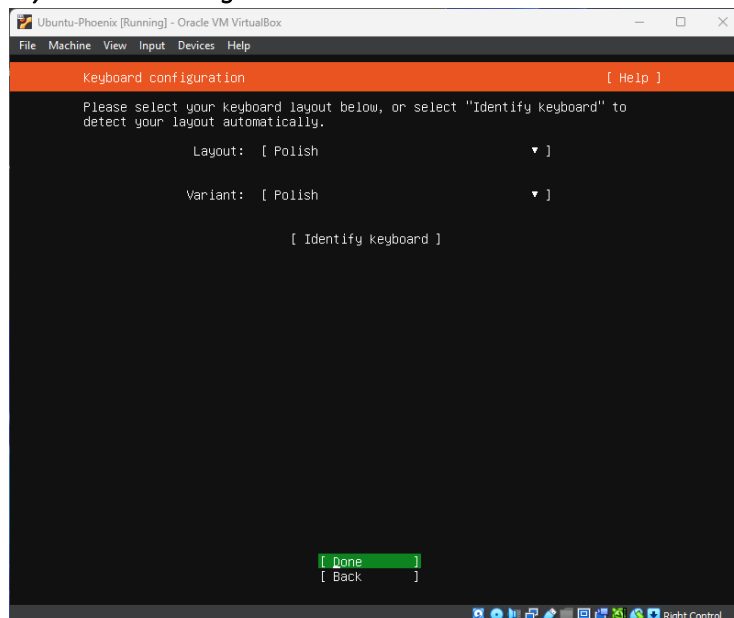
2.1.3.1 Language of the system: English



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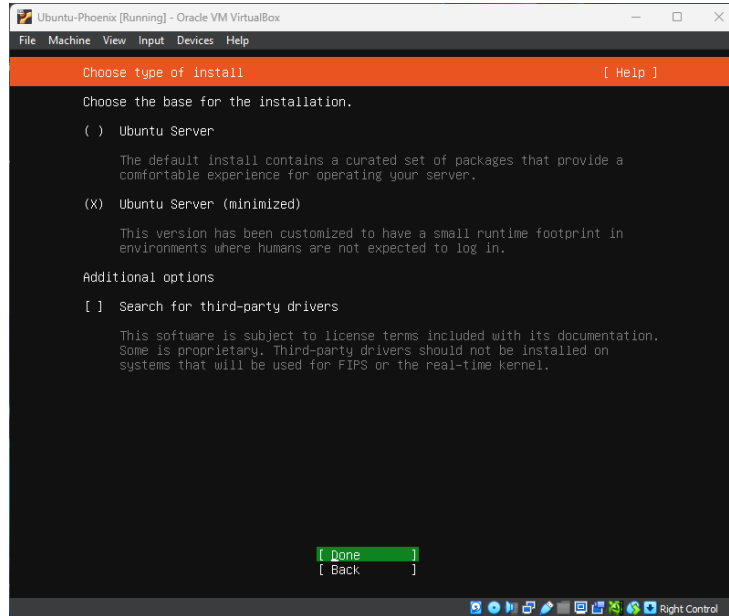


2.1.3.2 Keyboard Layout: Polis or English

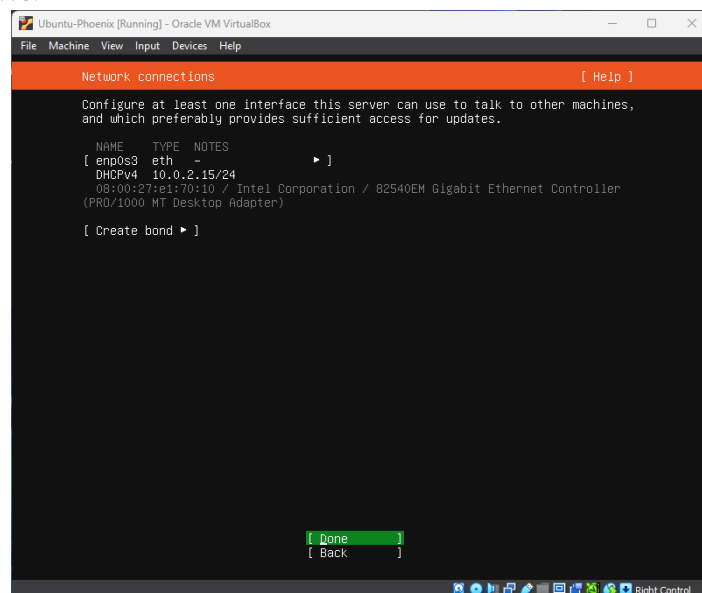


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2.1.3.3 *Install the minimum version of the server*

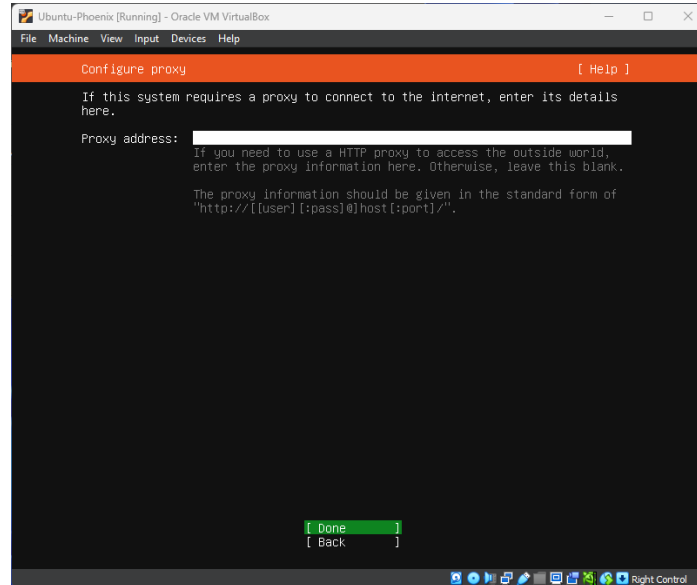


2.1.3.4 *IP Adres: DHCP*

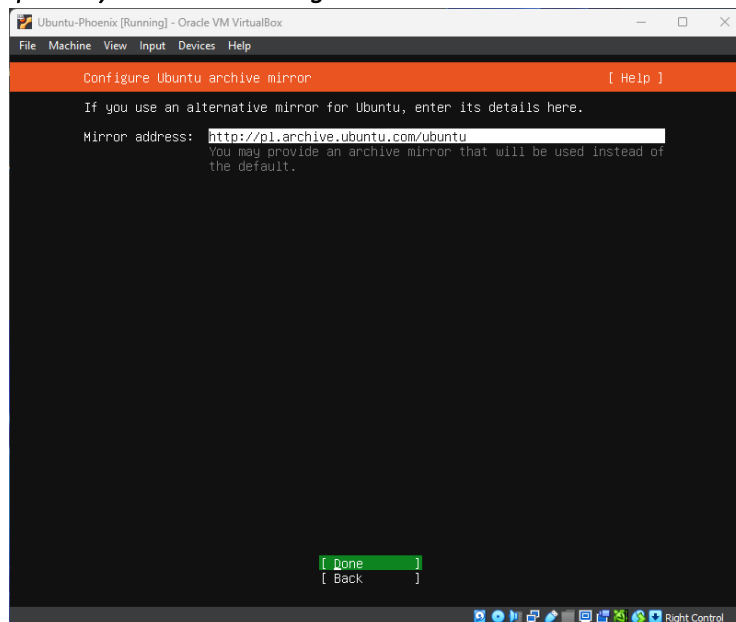


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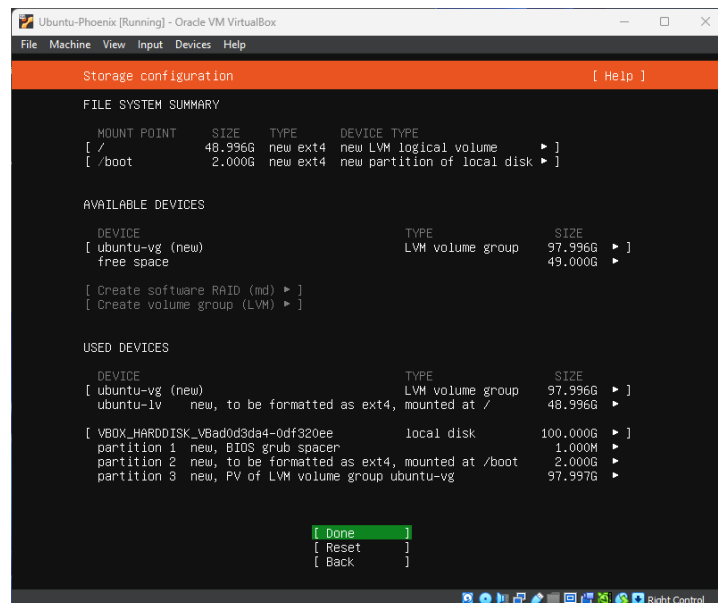
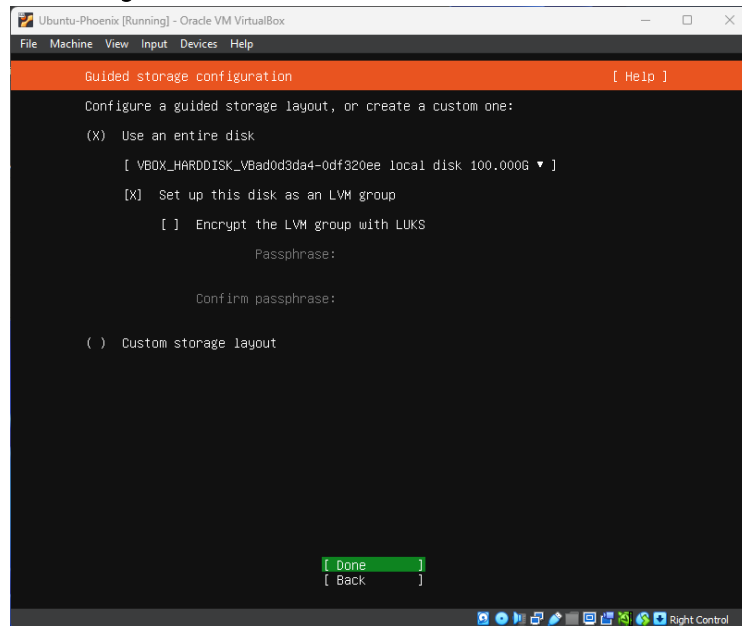
2.1.3.5 *Proxy: unchanged - none*



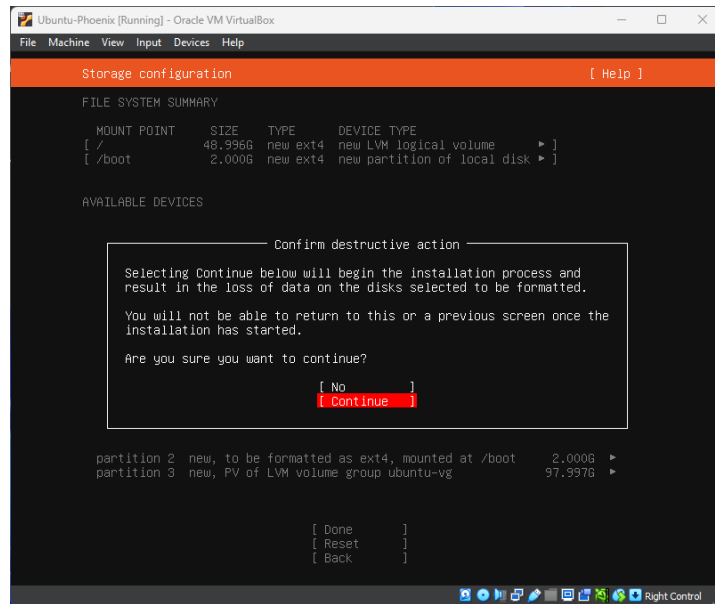
2.1.3.6 *Package repository address: unchanged*



2.1.3.7 Disk layout: unchanged - use an entire disk

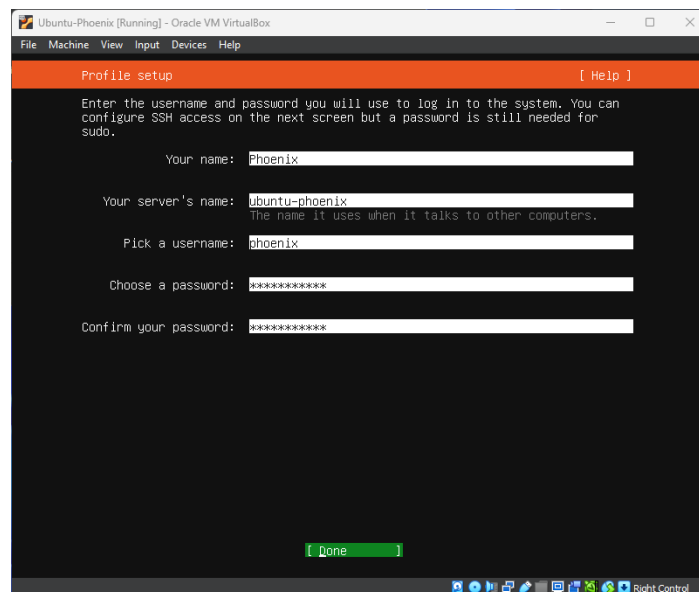


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2.1.3.8 Server name and username

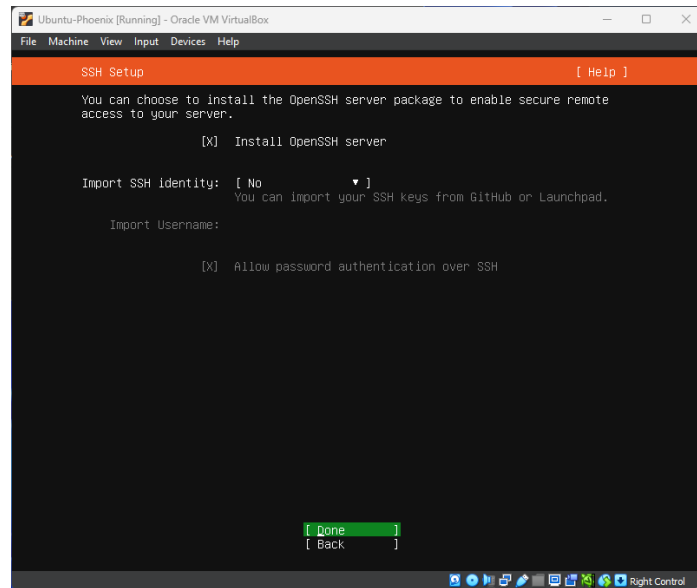
- Your name - descriptive name: Phoenix,
- Your server's name: ubuntu-phoenix
- Pick a username: phoenix
- Password: phoenixrtos



2.1.3.9 Installing an OpenSSH server

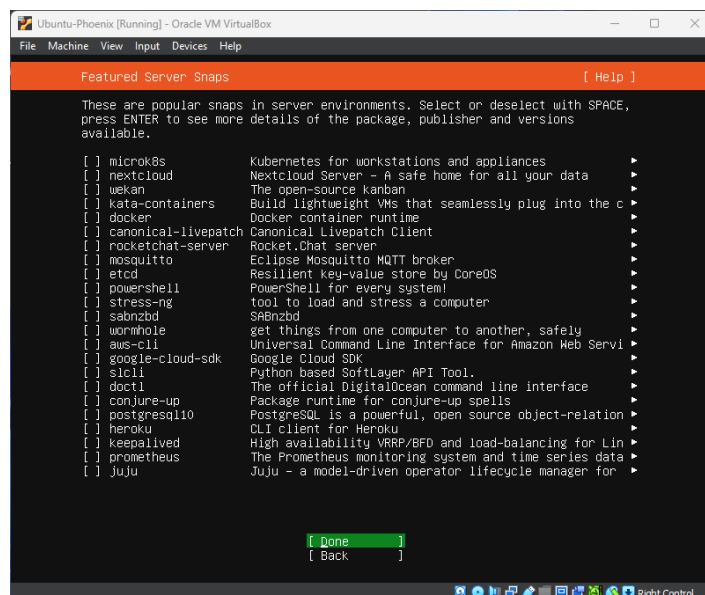
Check the option that you want to install an OpenSSH server.

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2.1.3.10 Installation of additional software

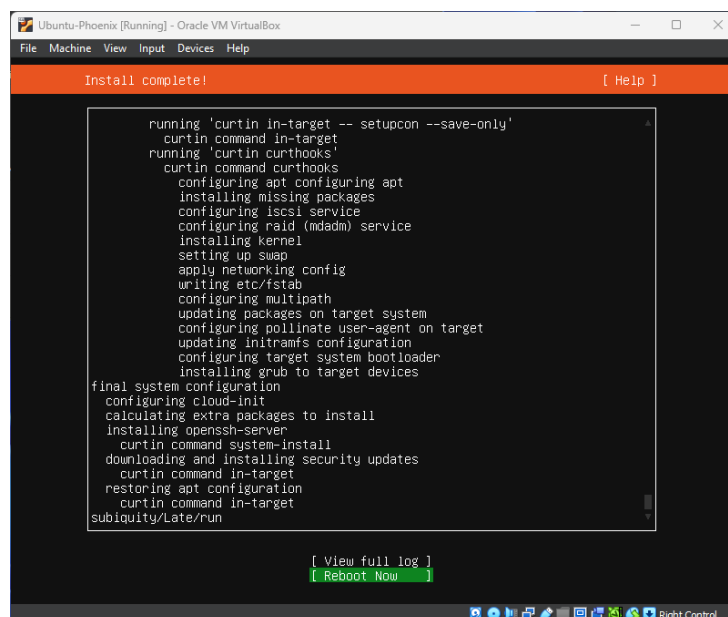
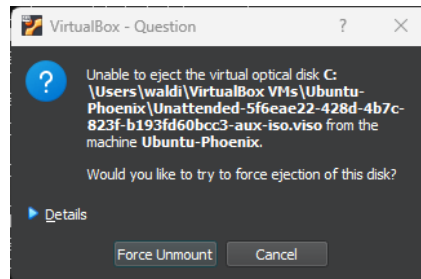
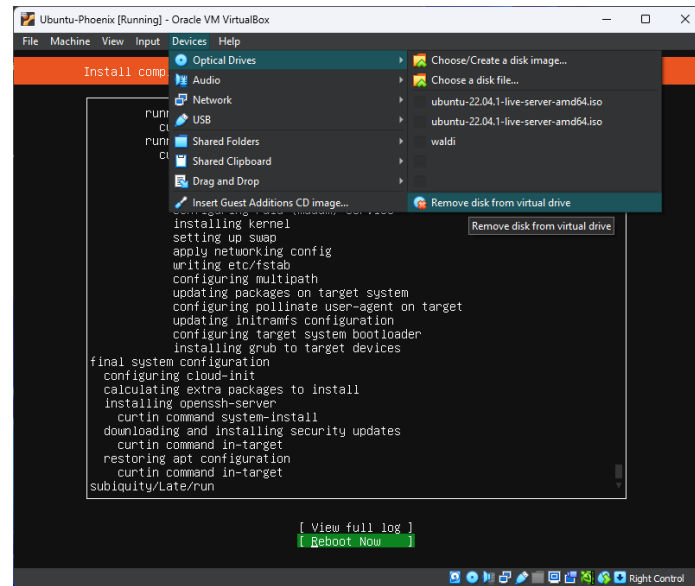
No changes, we don't install any additional packages.



2.1.3.11 Ubuntu has been installed

Note: Before rebooting the system, you must detach the installation disk image file in the VM configuration.

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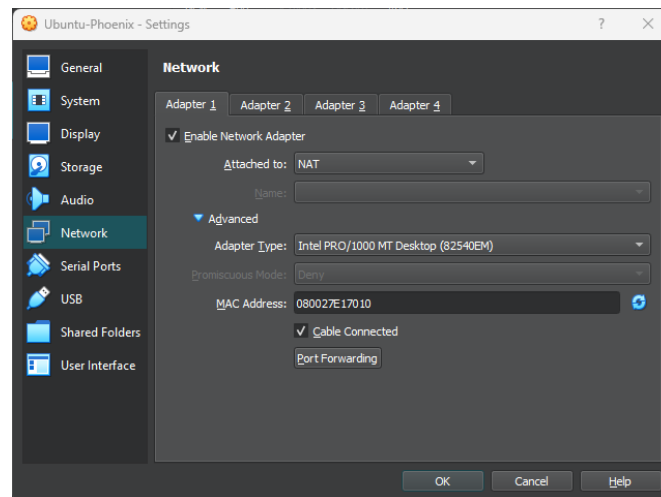


2.1.4 Port Forwarding - configuring SSH Access

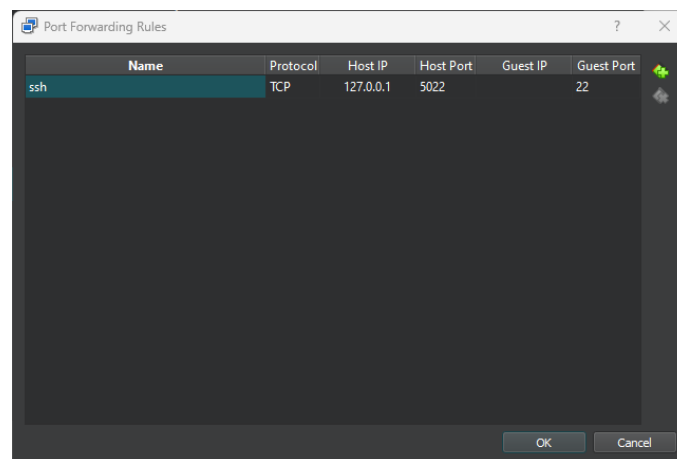
In the VM configuration, we add port forwarding to access the system using *the SSH protocol*

In the network settings of the virtual machine, select set port forwarding in the advanced settings.

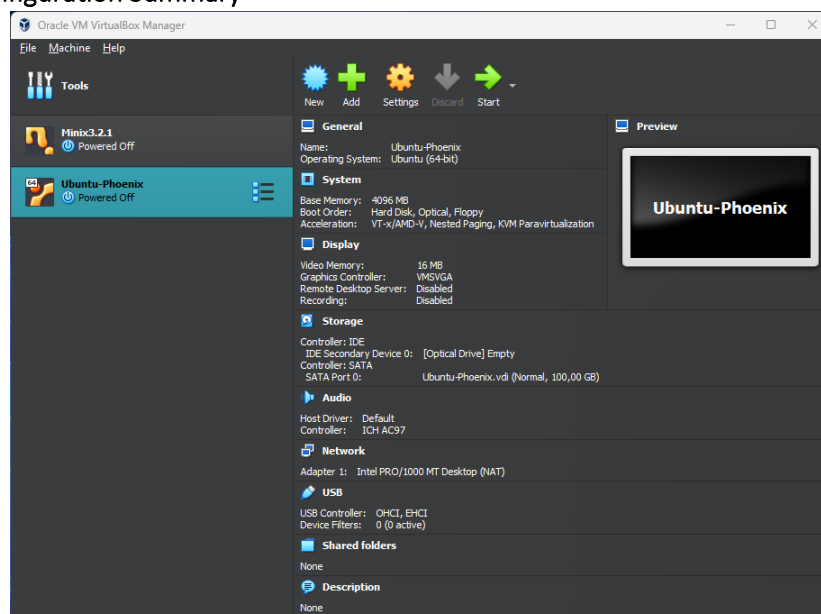
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We redirect port 5022 on the host system to port 22 on the guest system.



2.1.5 VM Configuration Summary



2.2 INSTALLATION IN WSL 2.0

Install *Ubuntu* (for *WSL 2.0*) from the *Windows Store*.

3 SOFTWARE INSTALLATION

Update information about available packages in the repository.

```
sudo apt update
```

Software update (optional, but recommended).

```
sudo apt upgrade
```

Using the *config.sh script*, we install the tools necessary to build the tools for building the Phoenix-RTOS system image.

Copy the *config.sh* file to the user's home directory.

If you are using a virtual machine under *VirtualBox*, copy the file using *the winscp* program. We connect to the local system on port 5022, which is redirected to port 22 of the guest system (*ubuntu-phoenix*).

If you are using *WSL*, you copy the file using a file explorer in *Windows*.

File *config.sh*

```
#!/bin/bash

# additional packets needed for toolchain compilation

apt-get update && \
# make sure debconf doesn't try to prompt (e.g. tzdata on Ubuntu)
DEBIAN_FRONTEND=noninteractive \
apt-get install -y --no-install-recommends \
    autoconf \
    automake \
    bzip2 \
    ca-certificates \
    g++ \
    gcc \
    libc6-dev \
    libglib2.0-dev \
    libmaxminddb-dev \
    libtool \
    make \
    patch \
    texinfo \
    unzip \
    wget \
    xz-utils \
    && rm -rf /var/lib/apt/lists/*

# phoenix-rtos build dependencies
# bcc, bin86: ia32 plo
# cpio: kernel
# hidapi: phoenixd
```

```
# mtd-utils: jffs2 image (imx6ull)
# genext2fs: ext2 image (ia32)
# xxd: stm32 image

apt-get update && \
# make sure debconf doesn't try to prompt (e.g. tzdata on Ubuntu)
  DEBIAN_FRONTEND=noninteractive \
  apt-get install -y --no-install-recommends \
    bcc \
    bin86 \
    cpio \
    genext2fs \
    git \
    libhidapi-dev \
    mtd-utils \
    xxd \
  qemu-system-x86 \
  && rm -rf /var/lib/apt/lists/*

# download qemu
apt-get update && \
  DEBIAN_FRONTEND=noninteractive \
  apt-get install -y --no-install-recommends \
  qemu\
  && rm -rf /var/lib/apt/lists/*
```

Setting execute permission for the *config.sh* file.

```
chmod u+x config.sh
```

Running the *config.sh* script as a root.

The *config.sh* script installs the necessary software. The guest system - ubuntu-phoenix, must have access to the internet.

```
sudo ./config.sh
```

In addition, you need to install *the cmake* program.

```
sudo apt install cmake
```

If you have problems installing packages, refresh the information about the available packages in the repositories and try the installation again.

```
sudo apt update
```

4 INSTALLATION OF ADDITIONAL SOFTWARE

We install useful software (optional) on the system.

```
sudo apt install nano
sudo apt install mc
```

5 BUILDING TOOLS FOR CREATING AN IMAGE OF THE PHOENIX-RTOS SYSTEM

The process of building an image of *the Phoenix-RTOS* system is described on <https://github.com/phoenix-rtos/phoenix-rtos-doc/blob/master/building/README.md>.

In the manual, I assume that we are working in the user's home directory `~/` and we are downloading the Phoenix-RTOS system sources to it.

CONSIDERATIONS FOR TOOL CONSTRUCTION IN THE LAB

Building tools takes quite a long time and if you use a network drive (your home directory is accessible via the network) it can take up to several hours. Therefore, if the tools will be built in a lab, the *toolchains* directory should be created on the local disk. Create a folder in *the /tmp* directory with a name like your username and create a *toolchains* directory in it. For *the wgr* user, the path to the *toolchains* directory will be */tmp/wgr/toolchains*.

5.1 DOWNLOADING THE SOURCE CODE FROM THE REPOSITORY

```
git clone --branch v3.1.0 --depth 1 --shallow-submodules --recursive  
https://github.com/phoenix-rtos/phoenix-rtos-project.git
```

We download the source code for version 3.1.0 of the Phoenix-RTOS system (shallow copy). The source code (and a copy of the repository) will be placed in *the ~/phoenix-rtos-project* directory.

5.2 BUILDING TOOLS FOR CREATING AN IMAGE OF THE PHOENIX-RTOS SYSTEM

Go to the *~/phoenix-rtos-project* directory with the source code you downloaded.

```
cd ~/phoenix-rtos-project
```

We build the required tools for the i386 architecture (the process takes: VirtualBox - 38 min, WSL 7 min.).

```
(cd phoenix-rtos-build/toolchain/ && ./build-toolchain.sh i386-pc-  
phoenix ~/toolchains/i386-pc-phoenix)
```

Once the tools are built, you can delete the *~/toolchains/i386-pc-phoenix/_build* directory containing the intermediate files.

```
rm -r ~/toolchains/i386-pc-phoenix/_build
```

CONSIDERATIONS FOR TOOL CONSTRUCTION IN THE LAB

In the lab for the *wgr* user, use the command (the *toolchains* directory on your local disk is in *the /tmp/wgr* directory):

```
(cd phoenix-rtos-build/toolchain/ && ./build-toolchain.sh i386-pc-  
phoenix /tmp/wgr/toolchains/i386-pc-phoenix)
```

5.3 ADDING THE PATH TO THE TOOLS DIRECTORY TO THE PATH ENVIRONMENT VARIABLE

Adding the path to the *~/toolchains/i386-pc-phoenix/i386-pc-phoenix/bin/* directory containing the executable files of the built tools to the *PATH* environment variable. We modify the *.profile* file in the user's home directory so that the value of the *PATH* variable is modified when the user logs in.

We add the following commands to the `~/.profile` file:

```
# set PATH so it includes $HOME/toolchains/i386-pc-phoenix/i386-pc-
# phoenix/bin if it exists
if [ -d "$HOME/toolchains/i386-pc-phoenix/i386-pc-phoenix/bin" ] ; then
    PATH="$HOME/toolchains/i386-pc-phoenix/i386-pc-phoenix/bin:$PATH"
fi
```

The PATH variable will be updated when the user logs in again.

CONSIDERATIONS FOR TOOL CONSTRUCTION IN THE LAB

In the lab for the `wgr` user, add the local disk directory `/tmp/wgr` to the path:

```
# set PATH so it includes /tmp/wgr/toolchains/i386-pc-phoenix/i386-
# pc-phoenix/bin if it exists
if [ -d "/tmp/wgr/toolchains/i386-pc-phoenix/i386-pc-phoenix/bin" ] ;
then
    PATH="/tmp/wgr/toolchains/i386-pc-phoenix/i386-pc-
    phoenix/bin:$PATH"
fi
```

6 CONFIGURATION AND IMAGE CONSTRUCTION OF THE PHOENIX-RTOS SYSTEM

6.1 DISABLE/ENABLE SYSTEM COMPONENTS BUILDING

We modify the configuration files to enable/disable the building of external components. We operate in the directory where the Phoenix-RTOS system sources have been downloaded - `~/phoenix-rtos-project`.

We enable *busybox* building, which gives us access to standard Linux tools (including the *ash shell*).

In order to reduce the time it takes to build a system image, we disable the construction of unnecessary external components. The *BOSYBOX* component is to be enabled (it is needed to start the telnet server).

File: `_projects/ia32-generic-qemu/build.project`

```
export PORTS_MBEDTLS=n
export PORTS_AZURE_SDK=n
```

File: `_targets/build.project.ia32-generic`

```
export PORTS_BUSYBOX=y
export PORTS_PCRE=n
export PORTS_OPENSSL=n
export PORTS_LIGHTTPD=n
export PORTS_DROPBEAR=n
export PORTS_LUA=n
export PORTS_LZO=n
export PORTS_OPENVPN=n
export PORTS_JANSSON=n
export PORTS_CURL=n
export PORTS_MICROPYTHON=n
```

6.2 CONFIGURING THE IP ADDRESS AND STARTING THE TELNET SERVER

We modify the configuration of the system so that the telnet server is started at startup. Telnet is sufficient for the needs of the laboratory (despite the lack of encryption) – it allows you to run several sessions (terminals) at the same time.

Modify the file (add the lines highlighted below) `_projects/ia32-generic-qemu/rootfs-overlay/etc/rc.psh`.

```
:{}:
export PATH=/bin:/sbin:/usr/bin:/usr/sbin
export HOME=/root
W /bin/bind devfs /dev
X /bin/posixsrv
X /sbin/lwip rtl:0x18:11
X /sbin/ifconfig en1 10.0.2.15
X /usr/sbin/telnetd
T /dev/console
X /bin/psh
```

6.3 CONSTRUCTION OF THE SYSTEM IMAGE

Go to the `~/phoenix-rtos-project` directory.

```
cd ~/phoenix-rtos-project
```

We build an image system.

```
TARGET=ia32-generic-qemu ./phoenix-rtos-build/build.sh clean all
```

Meaning of flags:

- *clean* - clean up the build directory,
- *all* – construction of all components,
- *core* - building basic programs for a given platform,
- *image* - creating the final image,
- *project* - building project-specific elements,
- *fs* - creating a file system.

A file `~/phoenix-rtos-project/_boot/ia32-generic-qemu/phoenix.disk` will be created as a disk image for the *Phoenix-RTOS* system. Building a system image takes about 1 minute.

7 STARTING THE PHOENIX-RTOS OPERATING SYSTEM

The Phoenix-RTOS system is started using the *qemu* program. We run the *qemu* program in a text-based environment.

```
qemu-system-i386 -cpu pentium3 -smp 1 -serial stdio -display curses
-drive file=~/phoenix-rtos-project/_boot/ia32-generic-
qemu/phoenix.disk,format=raw -netdev user,id=net0,hostfwd=tcp::5556-:23
-device rtl8139,netdev=net0
```

Port 5556 on the host system is redirected to port 23 (telnet) on the guest system.

In order to connect to the guest system, use the telnet command on the host system with the following parameters:

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```
telnet localhost 5556
```

The login details for the system are:

- username: root
- password: 1234

After running *qemu* in text mode, you can switch between the *qemu Esc+2* console and the *Esc+1 terminal* (*qemu* has problems reacting to this key sequence, you have to try several times).

In the *qemu* console, *quit* command closes *qemu*.